KDRSolvers:

Scalable, Flexible, Task-Oriented Krylov Solvers

PAW-ATM 2025
David K. Zhang, Rohan Yadav, Alex Aiken, Fred Kjolstad, Sean Treichler
Stanford University

$A\mathbf{x} = \mathbf{b}$ Done Right

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$$A\mathbf{x} = \mathbf{b}$$

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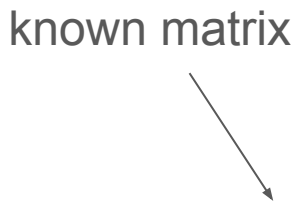
system of linear equations



$$Ax = b$$

system of linear equations

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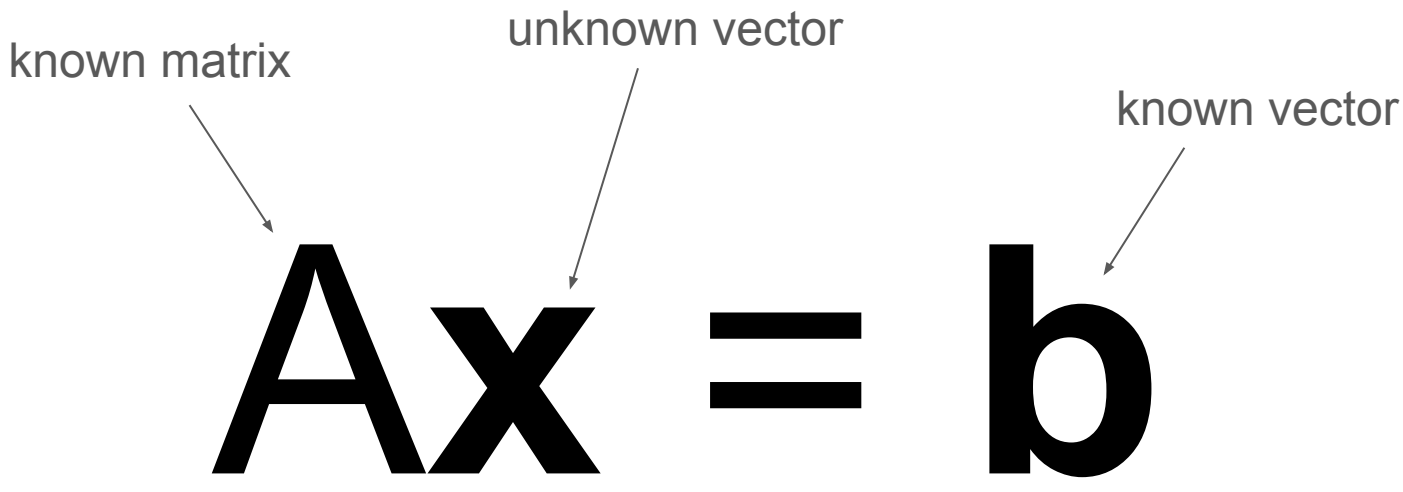

system of linear equations

A**x** = **b** is everywhere

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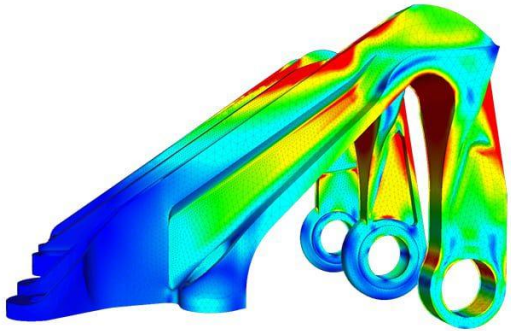

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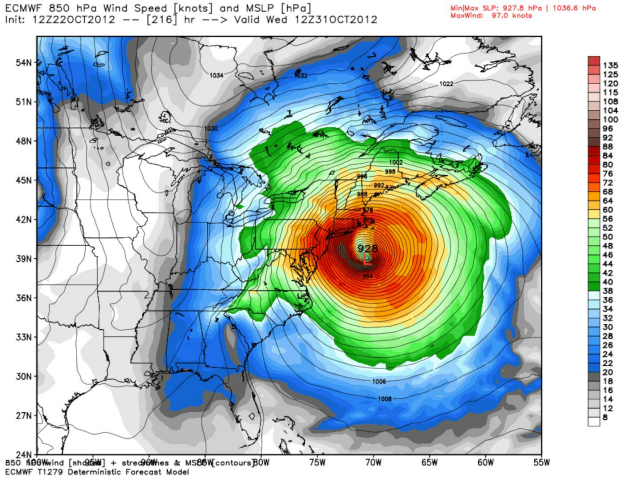

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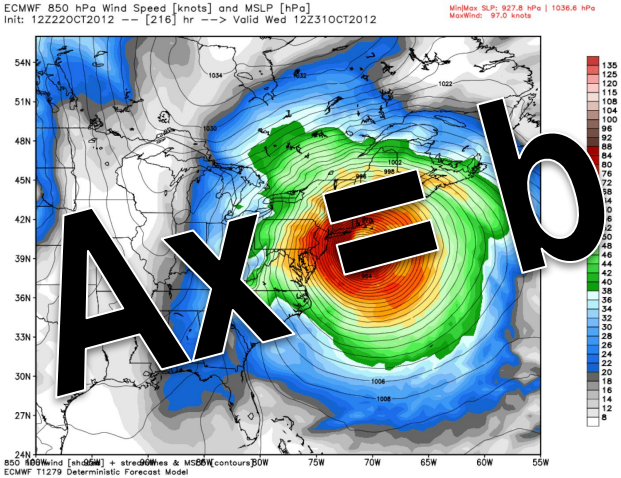

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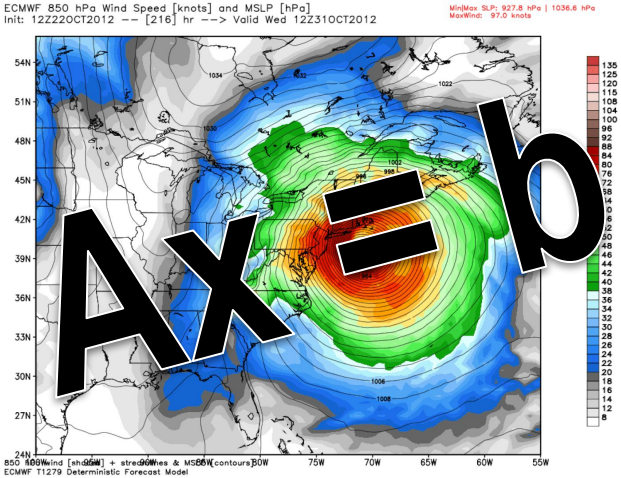

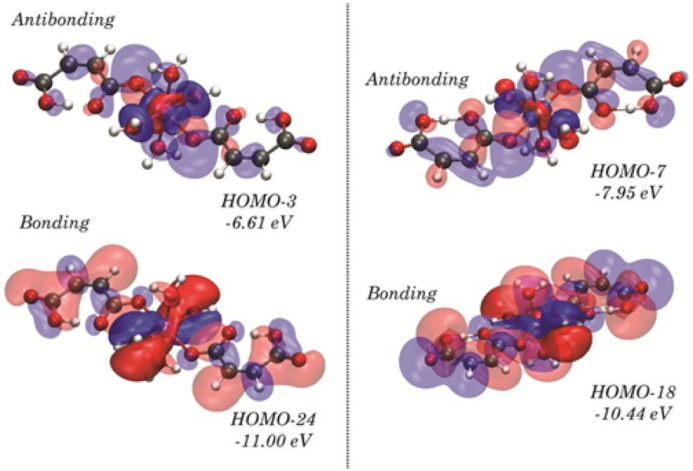

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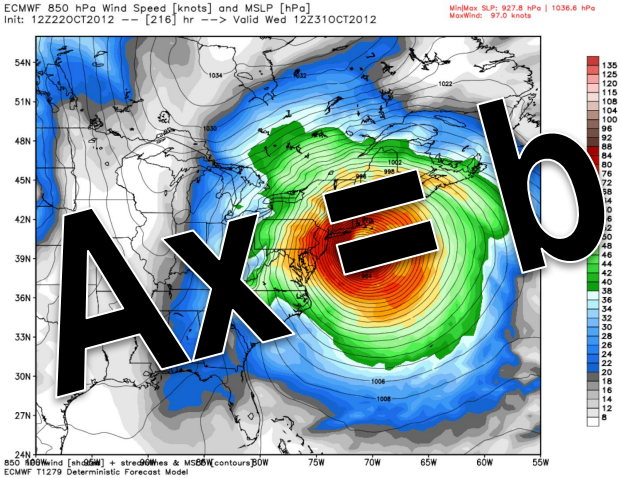

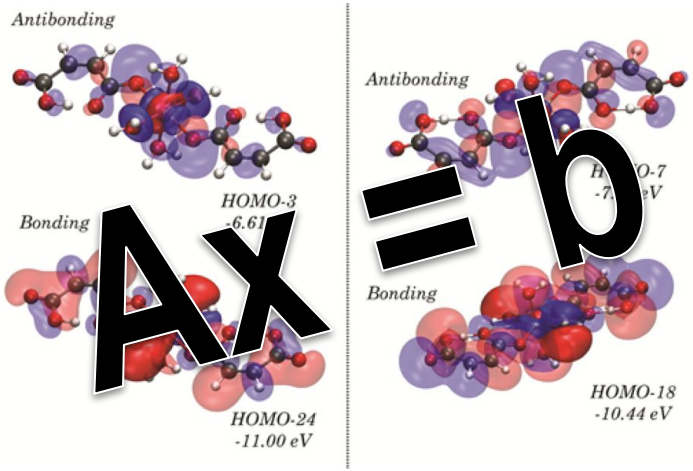

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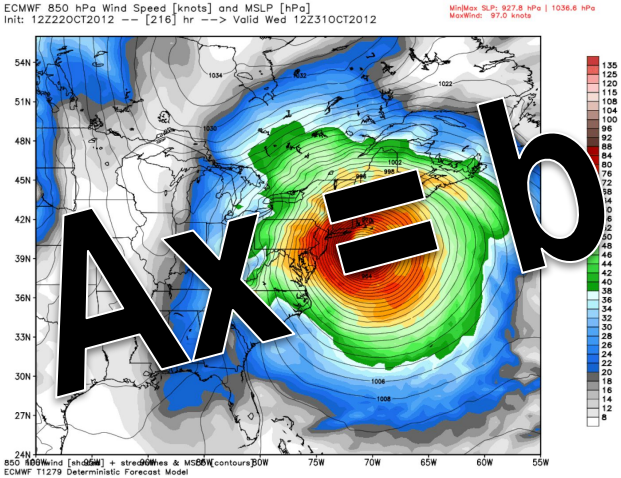

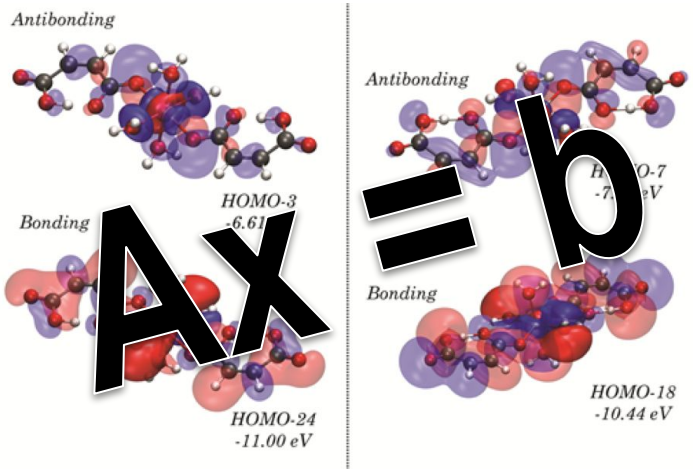

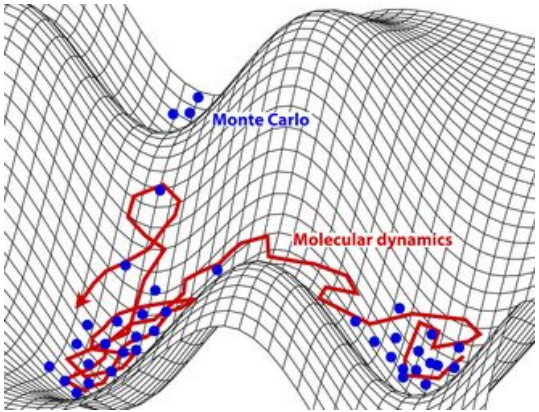

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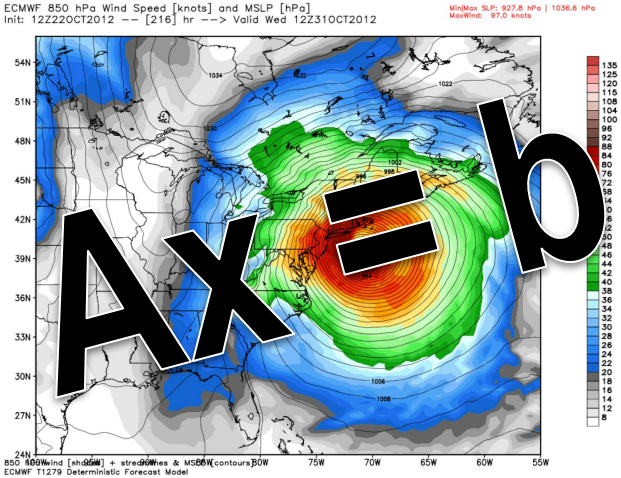

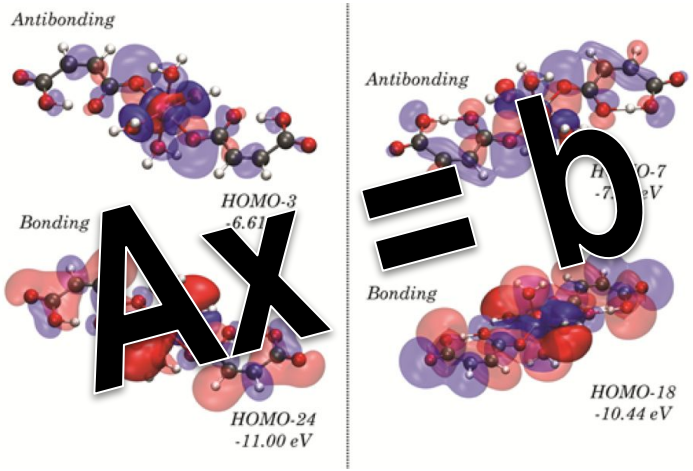

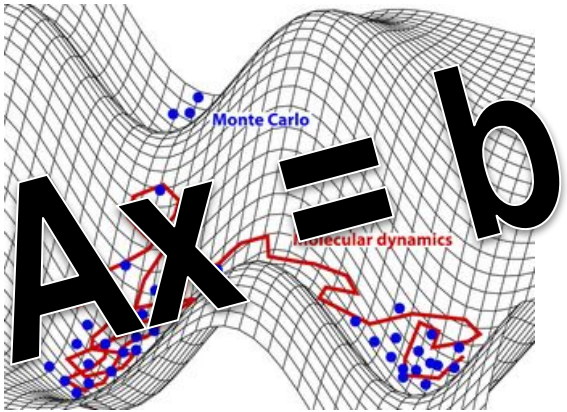

A**x** = **b** is a matter of national pride

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TOP 500

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TOP500

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	**El Capitan** - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581
2	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	**Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	**JUPITER Booster** - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	793.40	930.00	13,088
5	**Eagle** - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	

A**x** = **b** has been done before

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… but A**x** = **b** can be done **better**

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- **Parallel composition:**

  What if I want to interleave solving A**x** = **b** with other work?

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  What if my matrix A is stored in a format you don’t support?

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  What if I want to try out different data partitioning strategies?

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- **Layout flexibility: multi-operator systems**
  What if my matrix A lives in more than one place?

Sparse Matrix Storage Formats

Sparse Matrix Storage Formats

COO $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \\ 0 & 5 & 1 \end{bmatrix}$

entry	*row*	*col*
3	0	0
2	0	2
4	1	1
5	2	1
1	2	2

Sparse Matrix Storage Formats

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entry	*row*	*col*
3	0	0
2	0	2
4	1	1
5	2	1
1	2	2

CSR $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \\ 0 & 5 & 1 \end{bmatrix}$

entry	*col*
3	0
2	2
4	1
5	1
1	2

rowptr
[0, 1]
[2, 2]
[3, 4]

Sparse Matrix Storage Formats

COO $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \\ 0 & 5 & 1 \end{bmatrix}$

entry	*row*	*col*
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entry	*col*
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rowptr
[0, 1]
[2, 2]
[3, 4]

ELL $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \\ 0 & 5 & 1 \end{bmatrix}$

entry	
3	2
4	0
5	1

col	
1	3
2	0
2	3

Sparse Matrix Storage Formats

COO $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \\ 0 & & \end{bmatrix}$ **CSR** $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \end{bmatrix}$ **ELL** $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \\ & & 1 \end{bmatrix}$

Key Insight: all sparse matrix storage formats are defined by binary relations between three fundamental index spaces:

entry		
3		
2		
4	1	1
5	2	1
1	2	2

4	1
5	1
1	2

[3, 4]

5	1

	l
	3
	0
2	3

Sparse Matrix Storage Formats

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- Kernel index space *K*: nonzero entries of matrix

entry		
3		
2		
4	1	1
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Key Insight: all sparse matrix storage formats are defined by binary relations between three fundamental index spaces:

- Kernel index space *K*: nonzero entries of matrix
- Domain index space *D*: indices of input vector (columns)

entry		
3		
2		
4	1	1
5	2	1
1	2	2

4	1
5	1
1	2

[3, 4]

	3
	0
5	1
2	3

Sparse Matrix Storage Formats

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Key Insight: all sparse matrix storage formats are defined by binary relations between three fundamental index spaces:

- Kernel index space *K*: nonzero entries of matrix
- Domain index space *D*: indices of input vector (columns)
- Range index space *R*: indices of output vector (rows)

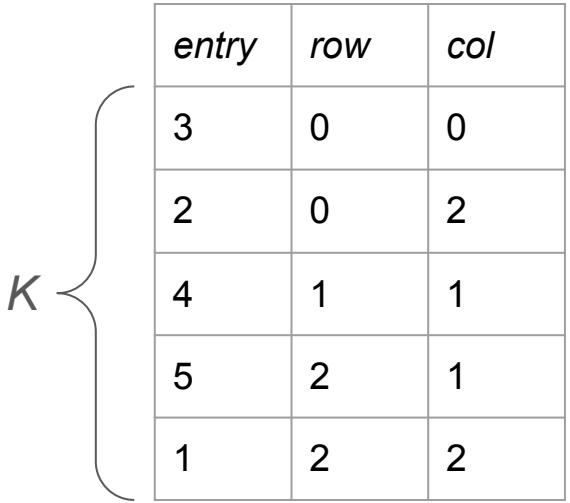
entry		
3		
2		
4	1	1
5	2	1
1	2	2

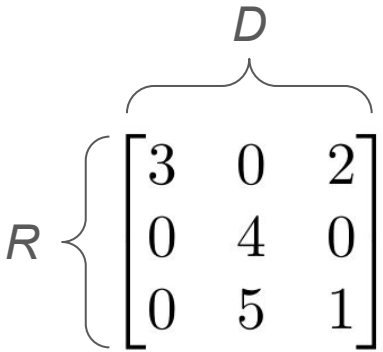
4	1
5	1
1	2

[3, 4]

	3
	0
5	1
2	3

COO Sparse Matrix Format





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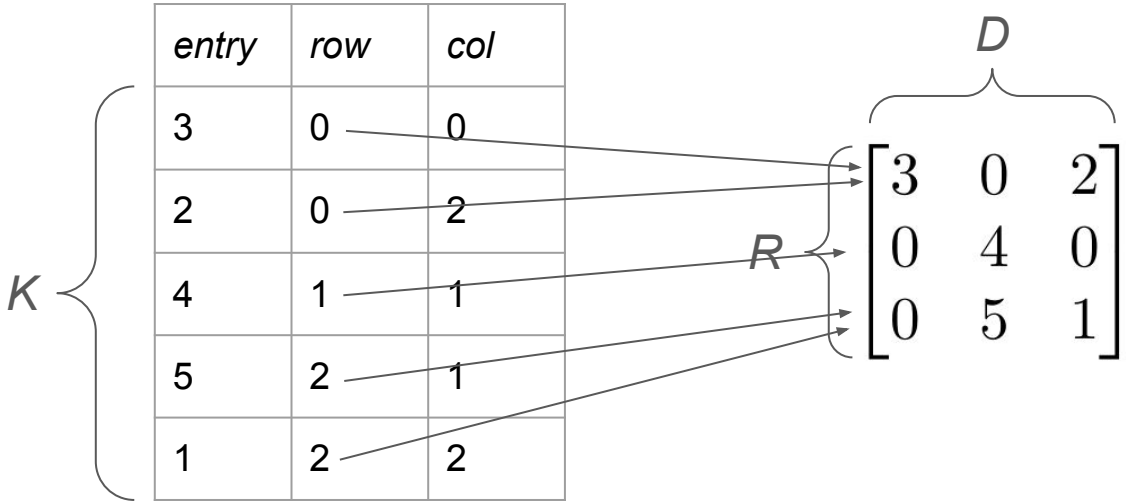

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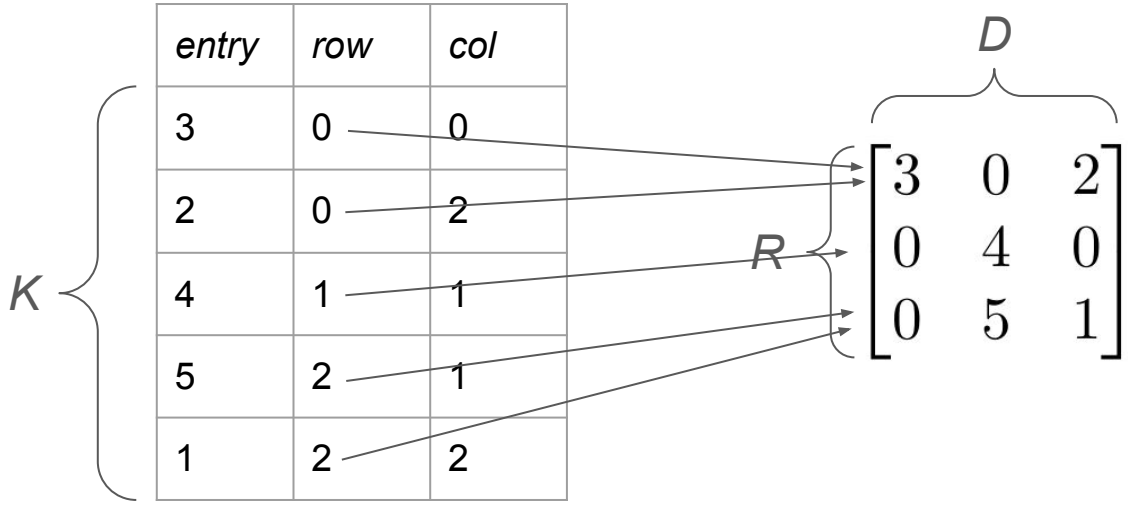

row: K → R

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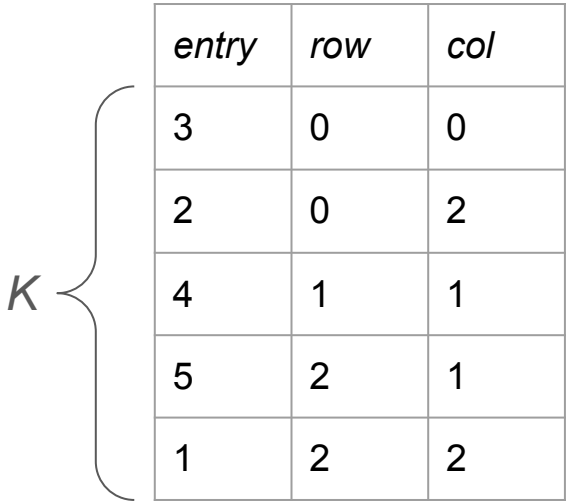

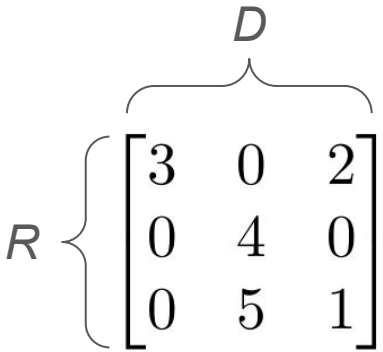

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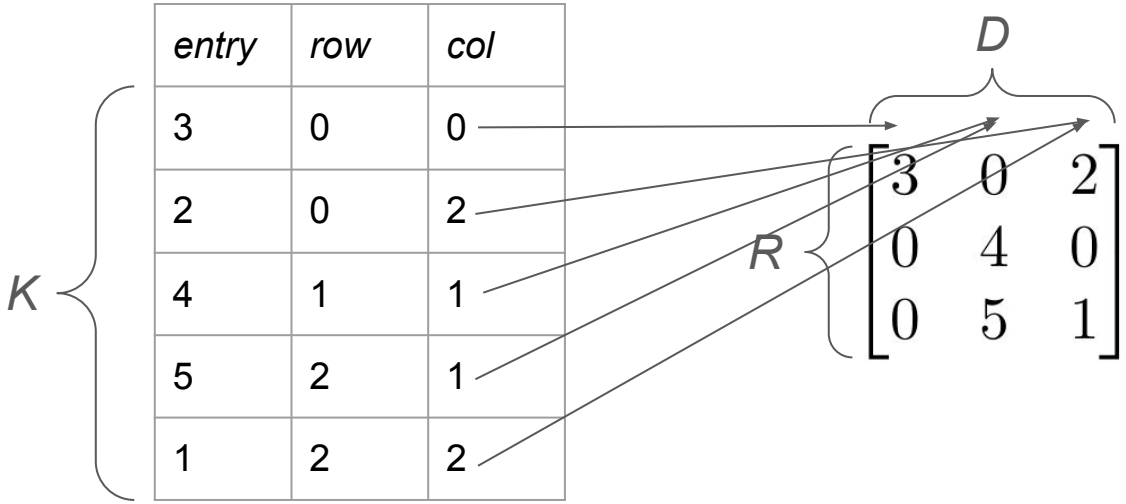

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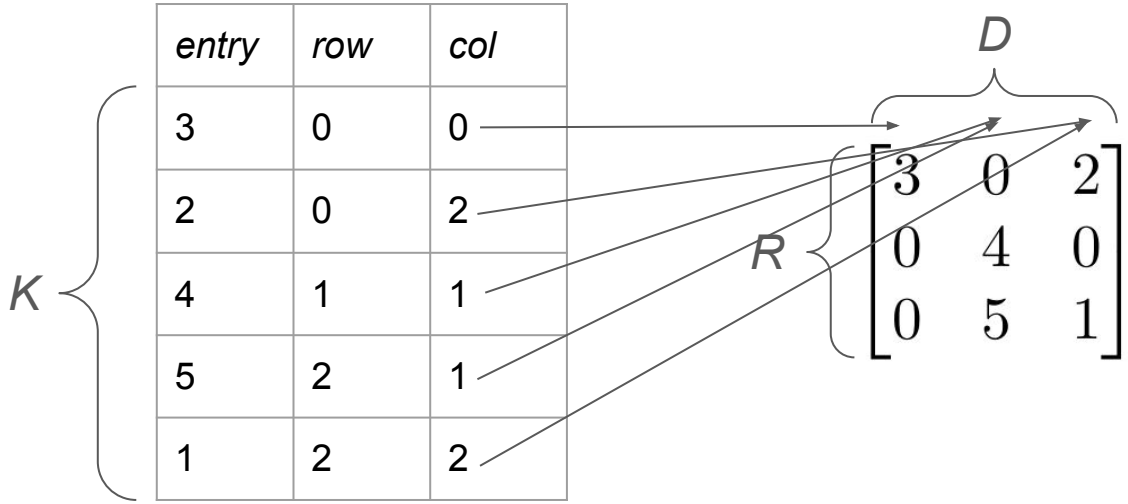


COO Sparse Matrix Format

K

entry	*row*	*col*
3	0	0
2	0	2
4	1	1
5	2	1
1	2	2

D

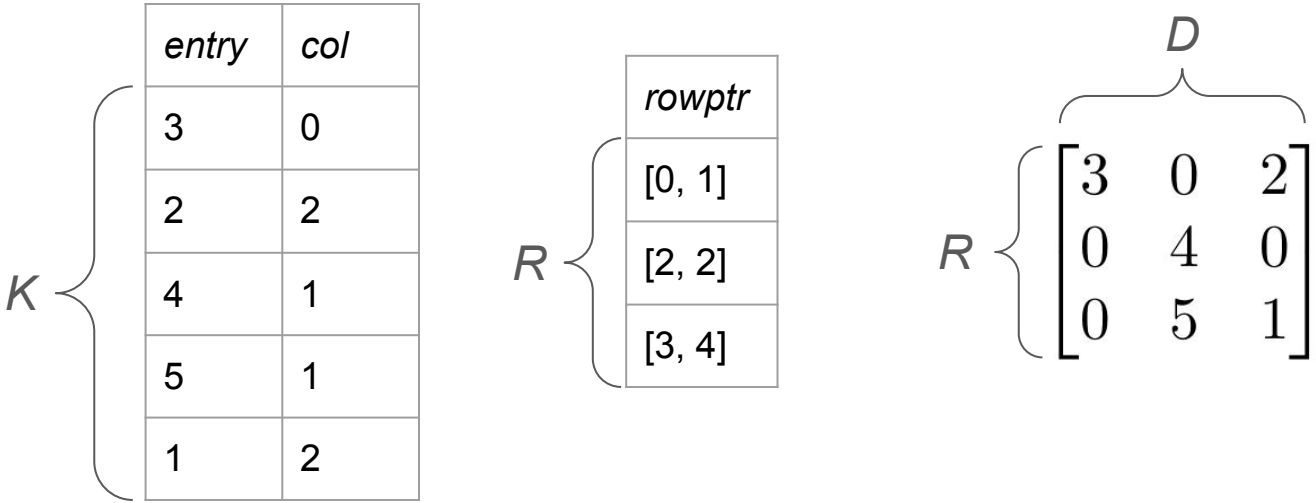
R

$$\begin{bmatrix} 3 & 0 & 2 \\ 0 & 4 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

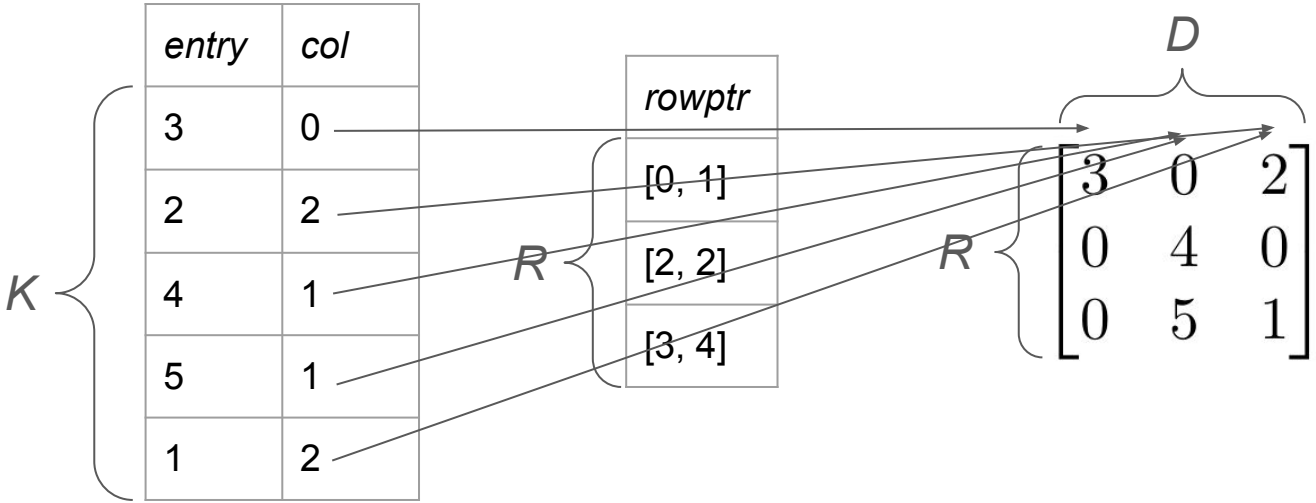
COO format is defined by:

- **column relation** *col*: $K \rightarrow D$
- **row relation** *row*: $K \rightarrow R$

CSR Sparse Matrix Format

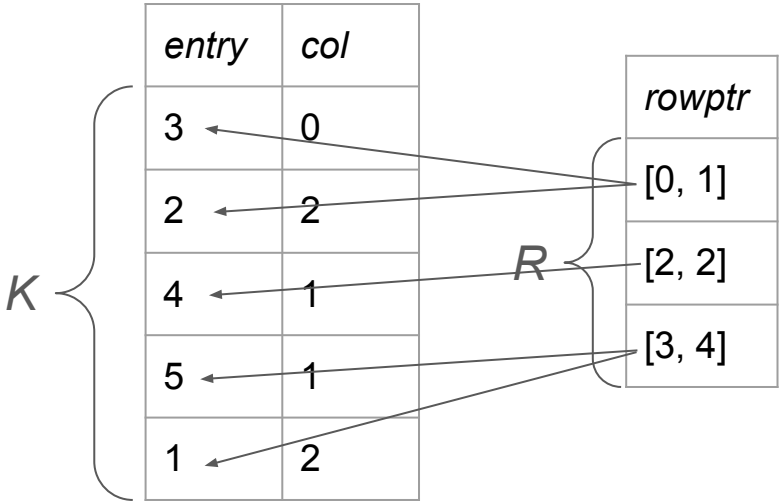


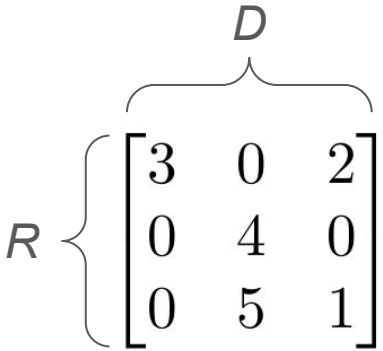
CSR Sparse Matrix Format



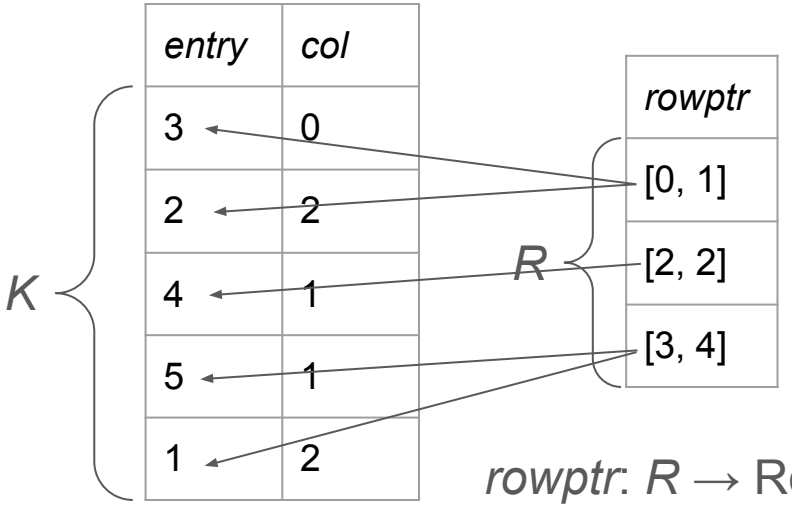
col: K → D

CSR Sparse Matrix Format

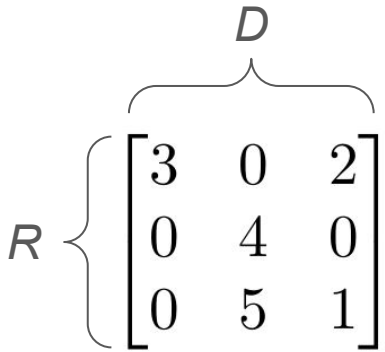




CSR Sparse Matrix Format



$rowptr: R \rightarrow \text{Rect}[K]$



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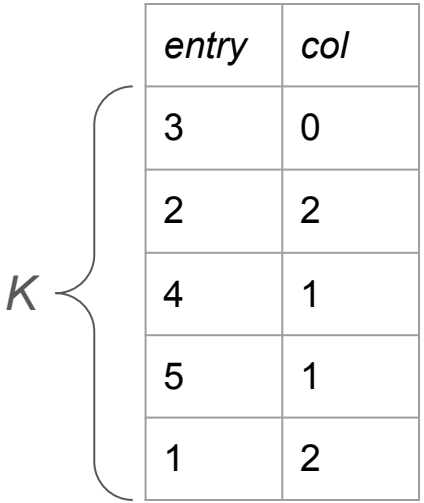
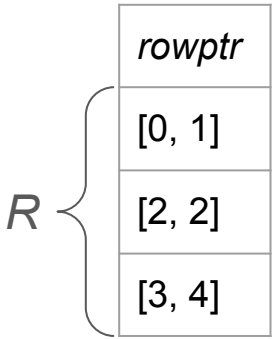
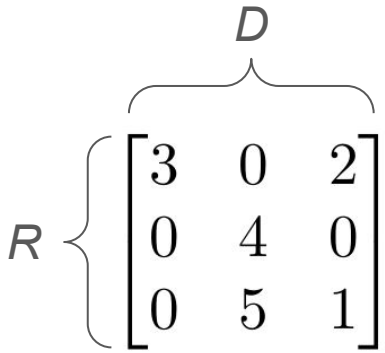


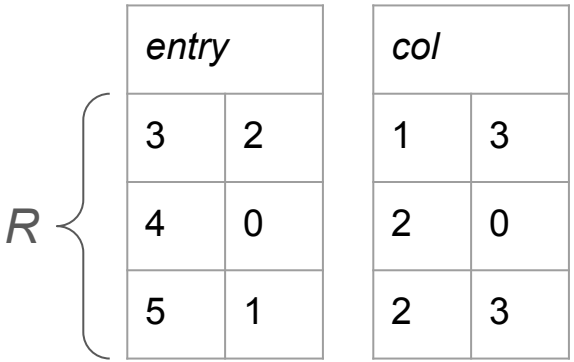


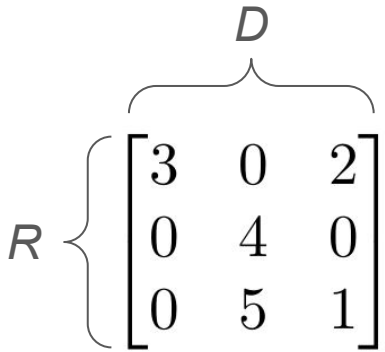


CSR format is defined by:

- **column relation** *col*: $K \rightarrow D$
- **row relation** *rowptr*: $R \rightarrow$ Rect[K]

ELL Sparse Matrix Format

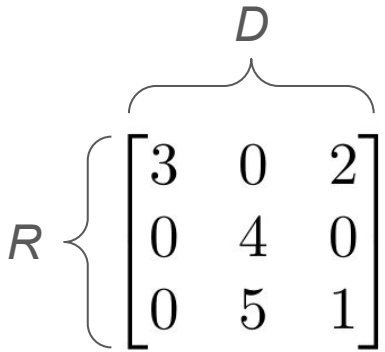




ELL Sparse Matrix Format

$K = R \times N$

	entry		*col*	
R	3	2	1	3
	4	0	2	0
	5	1	2	3



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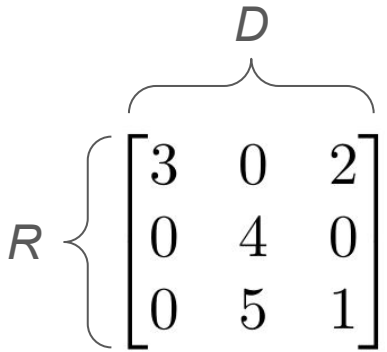

ELL format is defined by:

- column relation *col*: $K \rightarrow D$
- **implicit** row relation π_1: $R \times N \rightarrow R$

K, *D*, *R* Storage Formats

Format	Structural Assumptions	Column Relation	Row Relation
Dense	$K = R \times D$	$\pi_2 : R \times D \to D$ (implicit)	$\pi_1 : R \times D \to R$ (implicit)
COO	(none)	$\mathsf{col} : K \to D$	$\mathsf{row} : K \to R$
CSR	K is totally ordered	$\mathsf{col} : K \to D$	$\mathsf{rowptr} : R \to [K, K]$
CSC	K is totally ordered	$\mathsf{colptr} : D \to [K, K]$	$\mathsf{row} : K \to R$
ELL	$K = R \times K_0$	$\mathsf{col} : K \to D$	$\pi_1 : R \times K_0 \to R$ (implicit)
ELL′	$K = D \times K_0$	$\pi_1 : D \times K_0 \to D$ (implicit)	$\mathsf{row} : K \to R$
DIA	$D = \{1, \ldots, d\}$ $R = \{1, \ldots, r\}$ $K = K_0 \times \{1, \ldots, d\}$ $\mathsf{offset} : K_0 \to \mathbb{Z}$	$\mathsf{col} : (k_0, i) \mapsto i$ (implicit)	$\mathsf{row} : (k_0, i) \mapsto i - \mathsf{offset}(k_0)$ (implicit)
BCSR	$K = K_0 \times B_R \times B_D$ $D = D_0 \times B_D$ $R = R_0 \times B_R$ K_0 is totally ordered	$\mathsf{col} : K_0 \to D_0$	$\mathsf{rowptr} : R_0 \to [K_0, K_0]$
BCSC	$K = K_0 \times B_R \times B_D$ $D = D_0 \times B_D$ $R = R_0 \times B_R$ K_0 is totally ordered	$\mathsf{colptr} : D_0 \to [K_0, K_0]$	$\mathsf{row} : K_0 \to R_0$

Fig. 3. A wide variety of sparse matrix storage formats are realized in KDRSolvers as particular forms of column and row relations, subject to certain structural assumptions on the index spaces (K, D, R). Some column and row relations, marked as (implicit), can be specified without additional metadata as a consequence of structural assumptions or other auxiliary data. Here, π_i denotes the canonical projection function from a Cartesian product onto its ith coordinate, and whenever an index space K is assumed to be totally ordered, we denote by $[K, K]$ the set of contiguous intervals in K.

K, *D*, *R* Storage Formats

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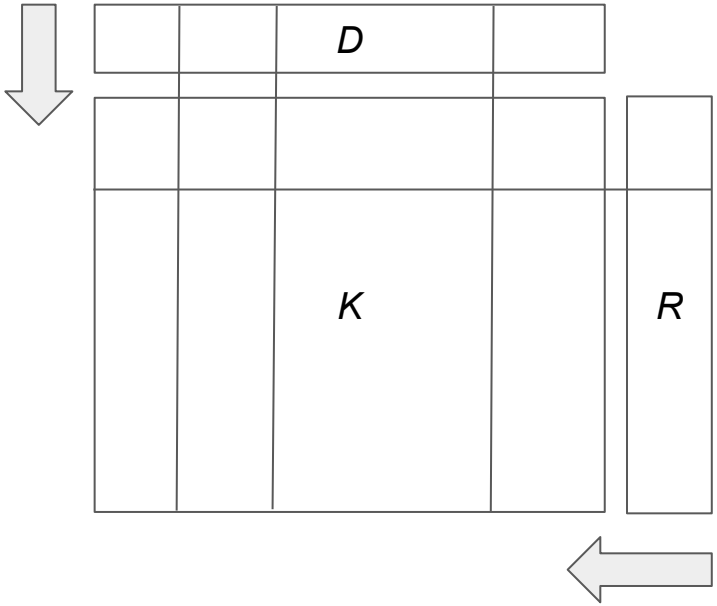
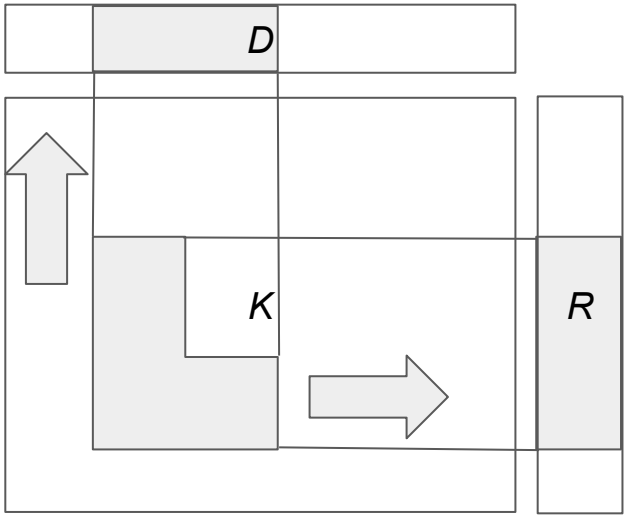
Then, we can do automatic **dependent partitioning**:

transfer a partition of one of (*K*, *D*, *R*) to the others by images/preimages

Matrix Dependent Partitioning Operators



Kernel partition from
domain and range partitions



Domain and range partitions
from kernel partition

Matrix Dependent Partitioning Operators

```
virtual Legion::IndexPartition
create_kernel_partition_from_domain_partition(
    Legion::IndexPartition domain_partition
) const = 0;

virtual Legion::IndexPartition create_kernel_partition_from_range_partition(
    Legion::IndexPartition range_partition
) const = 0;

virtual Legion::IndexPartition
create_domain_partition_from_kernel_partition(
    Legion::IndexSpace domain_space, Legion::IndexPartition kernel_partition
) const = 0;

virtual Legion::IndexPartition create_range_partition_from_kernel_partition(
    Legion::IndexSpace range_space, Legion::IndexPartition kernel_partition
) const = 0;
```

Multi-Operator Systems

Who says A**x** = **b** only involves a single matrix A and vector **b**?

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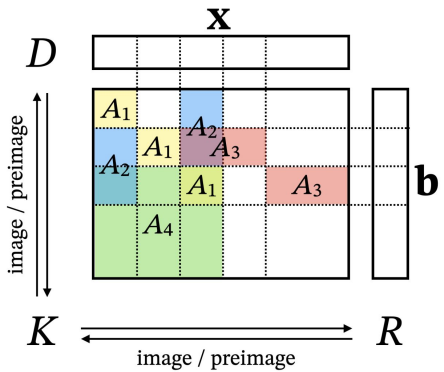

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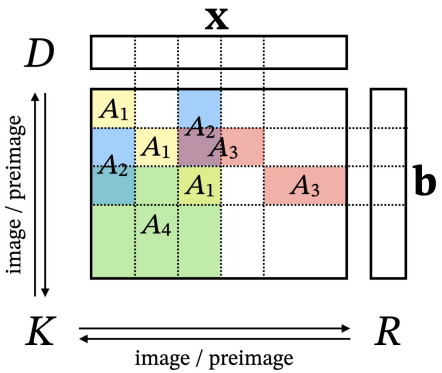
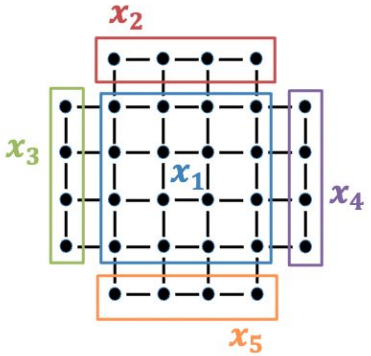
We allow a *logical* matrix or vector to consist of multiple *physical* pieces

Use data wherever it naturally exists in the context of your application





An example of ***heterogeneous dimension***: a 2D grid with four 1D boundary pieces. Note that we can easily enforce periodic boundary conditions by identifying $x_2 = x_5$ and $x_3 = x_4$.

Operations on Multi-Operator Systems

Vector operations (SCAL, AXPY, DOT) are done component-by-component

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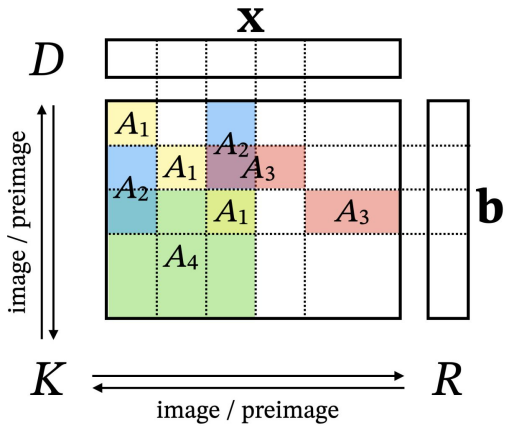

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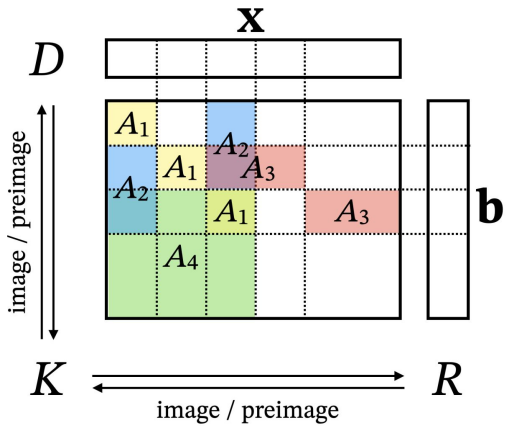

These are the only two operations needed
for **Krylov subspace methods**: CG, BiCG, MINRES, GMRES, …

Multiple Right-Hand Sides

$A\mathbf{x}_1 = \mathbf{b}_1$
$A\mathbf{x}_2 = \mathbf{b}_2$
$A\mathbf{x}_3 = \mathbf{b}_3$

$$\begin{bmatrix} A & & \\ & A & \\ & & A \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{x}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{bmatrix}$$

Related Systems

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$(A + \delta A_1)\,\mathbf{x}_1 = \mathbf{b}_1$
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Implementing Krylov Algorithms

```
template <typename ENTRY_T>
class CGSolver {

    \impname{}::Planner<ENTRY_T> &planner;
    static constexpr vec_id SOL = 0;
    static constexpr vec_id RHS = 1;
    vec_id P;
    vec_id Q;
    vec_id R;
    Scalar<ENTRY_T> res; // squared residual

public:

    explicit CGSolver(Planner<ENTRY_T> &planner)
        : planner(planner) {
        assert(planner.is_square());
        assert(!planner.has_preconditioner());
        P = planner.allocate_workspace_vector();
        Q = planner.allocate_workspace_vector();
        R = planner.allocate_workspace_vector();
        planner.copy(P, RHS);
        planner.copy(R, RHS);
        res = planner.dot(R, R);
    }

    void step() {
        planner.matmul(Q, P);
        Scalar<ENTRY_T> p_norm = planner.dot(P, Q);
        planner.axpy(SOL, res / p_norm, P);
        planner.axpy(R, -res / p_norm, Q);
        Scalar<ENTRY_T> new_res = planner.dot(R, R);
        planner.xpay(P, new_res, res, R);
        res = new_res;
    }

    Scalar<ENTRY_T> get_convergence_measure() const
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}; // class CGSolver<ENTRY_T>
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note: solver algorithms do not need to be aware of our new abstractions

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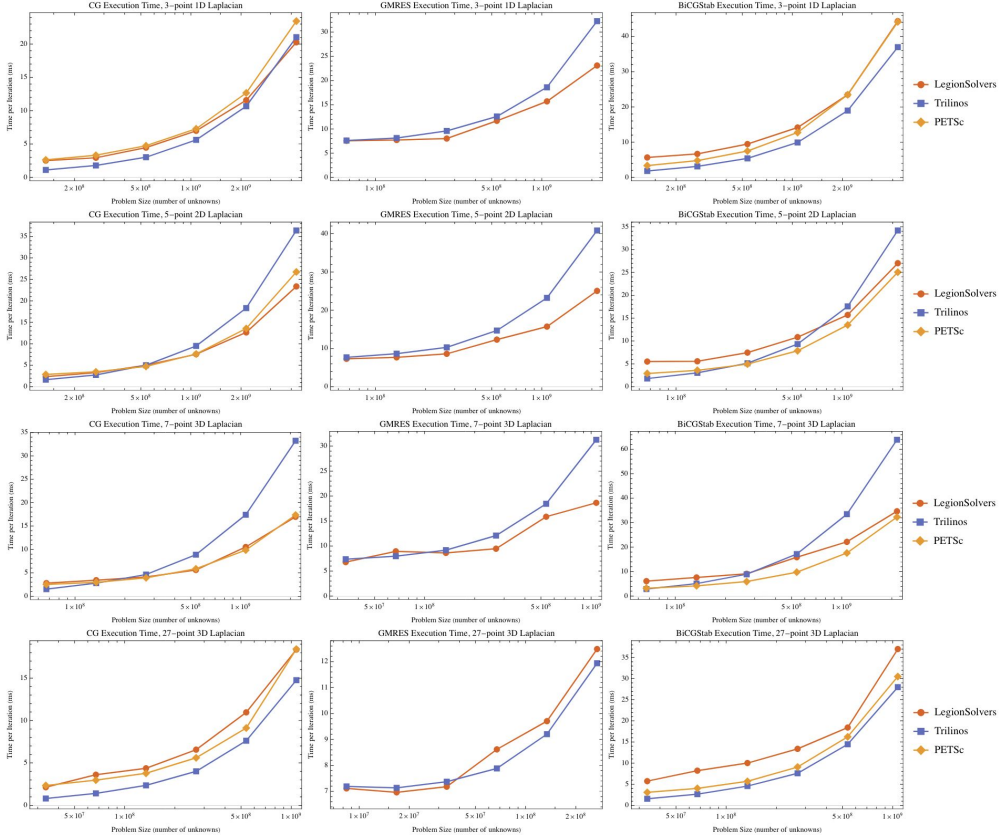
Benchmarks: Performance Comparison



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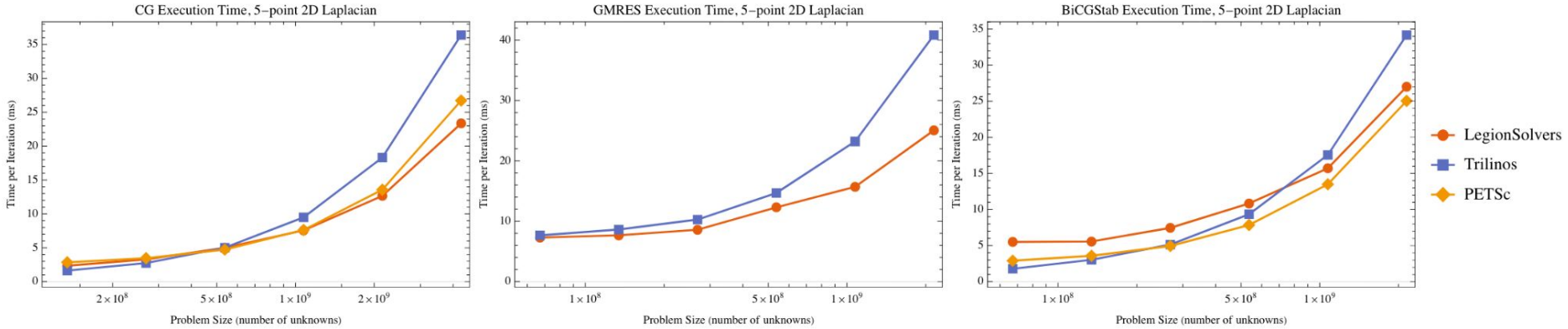

Benchmarks: Multi-Operator Systems

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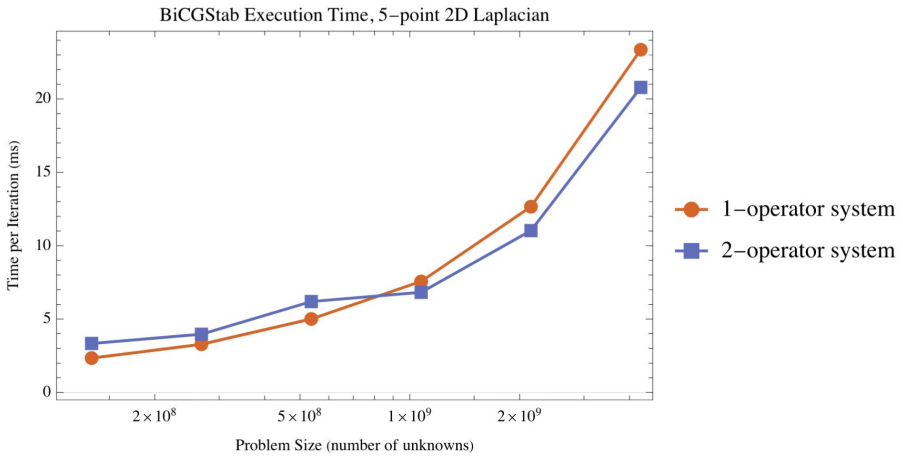


Figure 9: Execution time per iteration of BiCGStab using a 5-point Laplacian stencil on a $2^n \times 2^n$ grid, measured as a function of n, formulated as either a single-operator system (red) or a multi-operator system (blue).

Benchmarks: Dynamic Load-Balancing

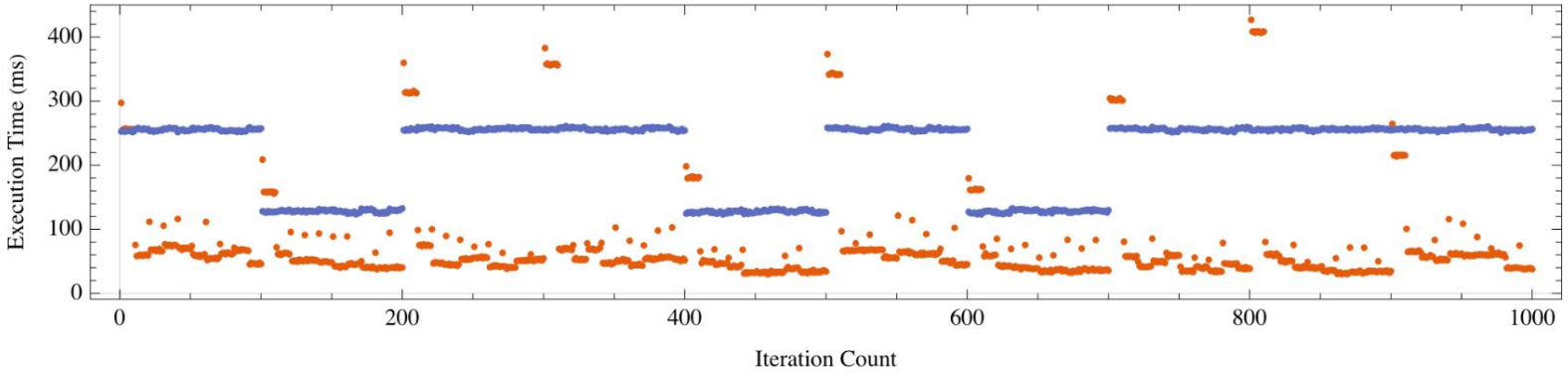
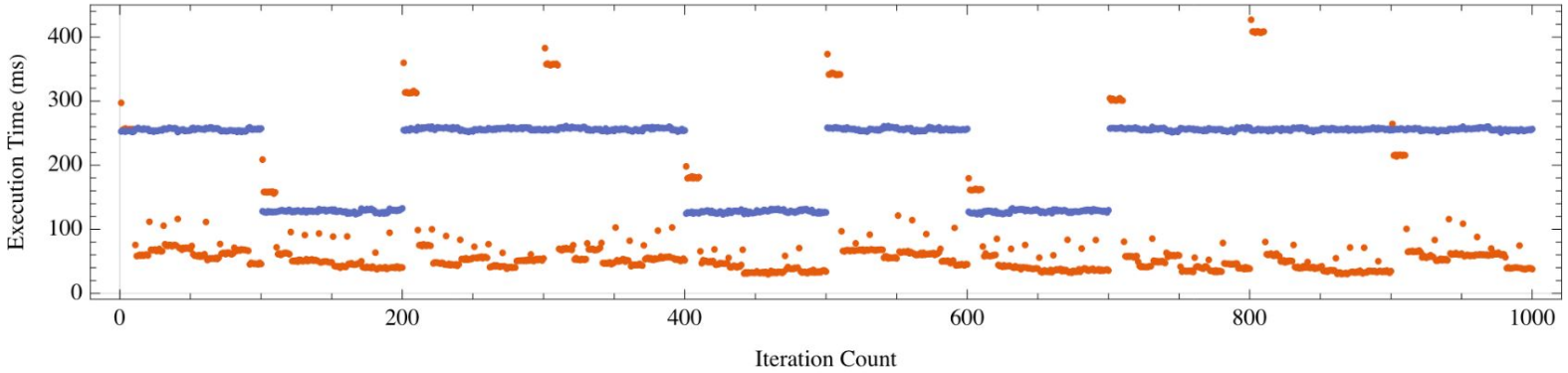


Figure 10: Execution time per iteration of BiCGStab using a 5-point Laplacian stencil on a $2^{16} \times 2^{16}$ grid with (red) and without (blue) dynamic load-balancing.

Benchmarks: Dynamic Load-Balancing



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In addition to a BiCGStab solve, each CPU also runs a dummy task that occupies a uniformly random number of its cores

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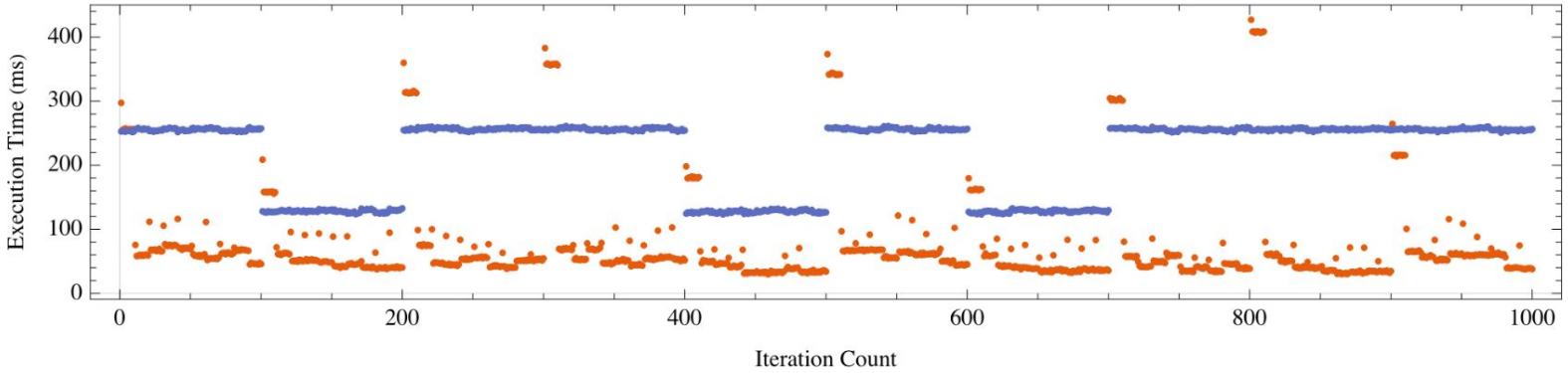

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Number of occupied cores is randomized every 100th CG iteration

Benchmarks: Dynamic Load-Balancing

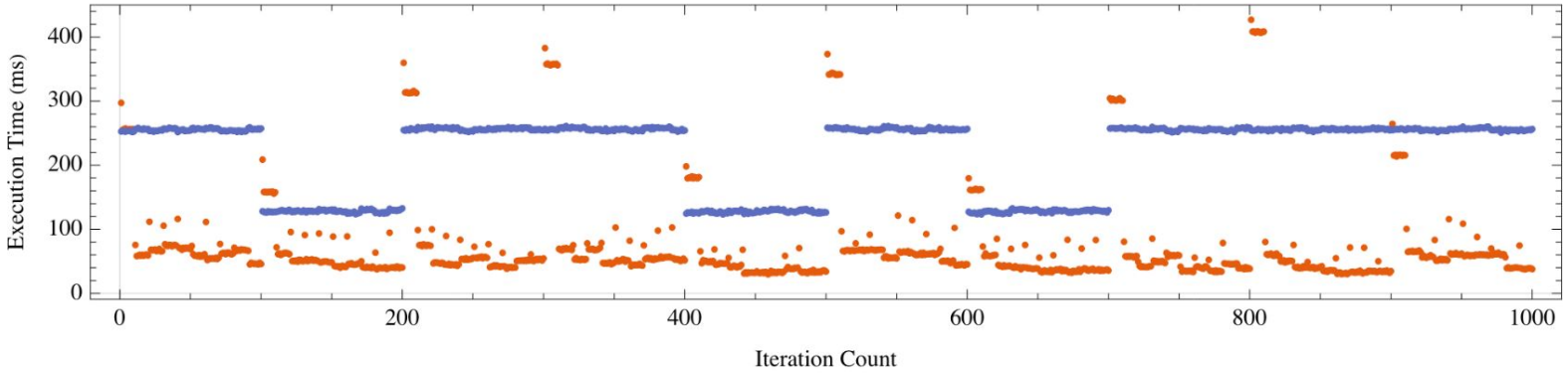

Figure 10: Execution time per iteration of BiCGStab using a 5-point Laplacian stencil on a $2^{16} \times 2^{16}$ grid with (red) and without (blue) dynamic load-balancing.

In addition to a BiCGStab solve, each CPU also runs a dummy task that occupies a uniformly random number of its cores

Number of occupied cores is randomized every 100th CG iteration

We run a dynamic load-balancing strategy every 10th CG iteration

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- Multi-operator systems **minimize communication** by processing data *in situ* without explicit copying or matrix/vector assembly steps.